

Simulation of Inelastic Neutron Scattering Spectra Directly from Molecular Dynamics Trajectories

Y. Q. Cheng,* A. I. Kolesnikov, and A. J. Ramirez-Cuesta

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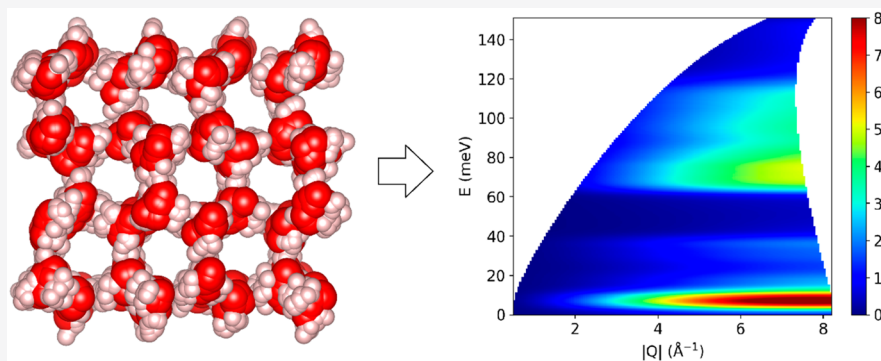
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ABSTRACT: Inelastic neutron scattering (INS) is a widely used technique to study atomic and molecular vibrations. With the increasing complexity of materials and thus the INS spectra, being able to simulate the spectra from various atomistic models becomes an essential step and also a major bottleneck for INS data analysis. The conventional approach using density functional theory and lattice dynamics often falls short when the materials of interest are complex (e.g., defective, disordered, heterogeneous, amorphous, large-scale), for which molecular dynamics driven by an interatomic force field is a more common approach. In this paper, we demonstrate a method to directly convert molecular dynamics trajectories into simulated INS spectra, including not only fundamental but also higher order excitations. The results are compared with data collected on various representative samples from different neutron spectrometers. This development will open great opportunities by providing the key tool to perform in-depth analysis of INS data and to validate and optimize computer models.

INTRODUCTION

Inelastic neutron scattering (INS) is a powerful tool to study phonons and molecular and atomic vibrations. It has been widely used in many fields such as chemistry, biology, soft matter, mineralogy, and condense matter physics.^{1–4} Among the many unique advantages of INS, an important but often overlooked one is the direct, quantitative, and theoretically rigorous relationship between the nuclear motion and the INS spectra.⁵ In other words, for nonmagnetic excitations, the atomic structure and dynamics are all we need to know to predict the peak positions and intensities in an INS spectrum. This is in stark contrast with other commonly used spectroscopic methods such as infrared and Raman scattering, where the complex photon-electron electromagnetic interactions dictate the spectra, and a theoretical prediction requires detailed understanding of the electronic structure, for which quantum models are required.⁶ To simulate an INS spectrum, however, one does not necessarily need a description of the electronic structure. A classical model with a decent empirical description of the interatomic interactions could suffice. The implication of this is significant, as it will allow accurate

modeling of INS spectra for much more complicated systems, systems for which quantum mechanical treatment is impractical.

In our previous papers,^{7,8} we have shown that accurate INS spectra can be obtained from full phonon information solved by lattice dynamics (LD), using the OCLIMAX program we developed. This approach, however, is only efficient for relatively small molecules or periodic unit cells with essentially ordered structure. For large or disordered systems, such as biomolecules, polymers, and glasses, the LD approach faces many challenges. First, it requires very accurate determination of the energy and interatomic forces, much more stringent than other types of calculation, slowing down the simulation. Second, LD calculations often take advantage of the symmetry

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of the crystal to significantly reduce the number of degrees of freedom. This is no longer possible for large and disordered systems. Third, if a system has many degrees of freedom, one must complete all in order to extract the dynamical matrix and therefore the phonon information; an incomplete calculation is futile. Fourth, diagonalization of the dynamical matrix is subject to larger errors when its size scales with the number of atoms in the system. To overcome these challenges, a potential solution is to use molecular dynamics (MD) simulations to study the vibrational behavior. MD simulations are widely performed with either quantum models using density functional theory (DFT) or classical models using empirical force fields. It has been routinely applied on many complex systems such as amorphous materials and large molecules.⁹ It does not require the level of accuracy as LD to obtain statistically meaningful results, and an incomplete or insufficient MD simulation (with fewer time-steps than desired) could still be used to simulate the spectra (albeit with larger noise at lower frequencies). An additional benefit of this method is that the anharmonic effects are naturally included. To convert the MD trajectories to phonon density of states (DOS), a well-established method is to take Fourier transformation of the velocity autocorrelation function.¹⁰ The DOS information can be further used to simulate INS spectra, as implemented in programs such as nMolDyn,¹¹ LiquidLib,¹² and MDANSE.¹³ However, these conversions are limited to fundamental excitations only, and there is no direct interface to allow users to simulate INS spectra for different neutron scattering instruments.

In this paper, we take advantage of the already established LD theory and interface in OCLIMAX⁷ and demonstrate a method to convert the MD trajectory information into useful phonon information, followed by full INS spectra simulations using OCLIMAX. We will first discuss the theory. We will then discuss several examples, including ice Ih, a metal–organic framework ZIF-8, and silica glass. We will show how this new development allows us to obtain more information than what was previously achieved, as well as its great potential for broader applications.

THEORETICAL BACKGROUND

Under incoherent approximation, the INS spectra can be calculated following⁵

$$S_{\text{inc}\pm 1}(\mathbf{Q}, \omega) = \sum_d \frac{\sigma_d}{2M_d} \exp(-2W_d) \sum_s \frac{|\mathbf{Q} \cdot \mathbf{e}_{ds}|^2}{\omega_s} \left(n_s + \frac{1}{2} \pm \frac{1}{2} \right) \delta(\omega \mp \omega_s) \quad (1)$$

where σ_d and M_d are the neutron scattering cross section and the atomic mass of atom d , respectively, ω_s is the frequency (of mode s), $\mathbf{e}_{ds}$ is the polarization vector, W_d is the Debye–Waller factor, and n_s is the population of phonon mode s following the Bose–Einstein statistics. Compared to the full equation including coherent scattering, this is a great simplification but still preserves high accuracy for many systems of interest. The incoherent approximation produces sufficiently accurate results if 1) the sample contains a significant amount of incoherent scatterers such as hydrogen or 2) the wavevector coverage, $4\pi Q^2 \Delta Q$ (where Q is neutron momentum transfer, and ΔQ is momentum transfer resolution), is significantly larger than the Brillouin zone volume. Condition 1) is easily met in many molecular/organic systems where hydrogen is abundant. Even for a material that only contains strong coherent scatterers, condition 2) is usually true when the

crystal structure is relatively complex with a large unit cell and therefore a small Brillouin zone, or the structure is disordered or amorphous and the true “unit cell” is essentially the size of the entire system.¹ To be more specific, the size of the Brillouin zone is about $2\pi/a$ where a is the size of the unit cell. This means for a crystal with a lattice constant greater than 6 Å, the effect of coherent scattering is expected to be rather weak at $Q > 2 \text{ Å}^{-1}$ even if all elements are coherent scatterers. For the above reasons, and because we are mostly interested in applications in chemistry, biology, and soft matter, a coherent effect is rarely a concern.

To further simplify eq 1, we define the displacement tensor⁷

$$\mathbf{B}_{ds} = \mathbf{e}_{ds} \mathbf{e}_{ds}^T = \begin{bmatrix} e_{ds1}e_{ds1} & e_{ds1}e_{ds2} & e_{ds1}e_{ds3} \\ e_{ds2}e_{ds1} & e_{ds2}e_{ds2} & e_{ds2}e_{ds3} \\ e_{ds3}e_{ds1} & e_{ds3}e_{ds2} & e_{ds3}e_{ds3} \end{bmatrix}$$

and

$$\mathbf{A}_d = \sum_s \mathbf{B}_{ds} (2n_s + 1) \quad (2)$$

Unless the system of interest is strongly anisotropic, eq 1 can be reduced to

$$S_{\text{inc}\pm 1}(\mathbf{Q}, \omega) = \sum_d \frac{\sigma_d}{2M_d} \exp(-2W_d) \sum_s \frac{Q^2 \text{tr}(\mathbf{B}_{ds})}{\omega_s} \left(n_s + \frac{1}{2} \pm \frac{1}{2} \right) \delta(\omega \mp \omega_s) \quad (3)$$

where

$$W_d = -\frac{1}{3} Q^2 [\text{tr}(\mathbf{A}_d)] \quad (4)$$

When taking the LD approach, the information on all phonon modes is available, thus the displacement tensor can be obtained for a full solution of eq 3. However, this is not the case when an MD trajectory is the only input. From the velocity autocorrelation function of the MD trajectory¹⁰

$$C_d(t) = \left\langle \frac{\mathbf{v}_d(t + t_0) \cdot \mathbf{v}_d(t_0)}{\mathbf{v}_d(t_0) \cdot \mathbf{v}_d(t_0)} \right\rangle \quad (5)$$

The partial phonon DOS for each atom can be obtained:

$$g_d(\omega) = \frac{1}{2\pi} \int \exp(-2i\pi\omega t) C_d(t) dt \quad (6)$$

A key relationship connecting the LD phonons and the MD trajectory is

$$g_d(\omega) d\omega = \sum_{\omega < \omega_s < \omega + d\omega} \frac{\text{tr}(\mathbf{B}_{ds})}{3} \quad (7)$$

which translates to another expression of eq 3:⁵

$$S_{\text{inc}\pm 1}(\mathbf{Q}, \omega) = \sum_d \frac{3\sigma_d}{2M_d} \exp(-2W_d) \frac{Q^2 g_d(\omega)}{\omega} \left(n + \frac{1}{2} \pm \frac{1}{2} \right) \quad (8)$$

Obviously, if the displacement tensor is known from the LD simulation, one can always derive the atom-specific phonon DOS. However, if the atom-specific phonon DOS is obtained from an MD trajectory, it is not possible to extract all phonon information for each mode. What can be extracted, instead, is the “density” for the trace of the displacement tensor, binned within the specific energy range. In other words, it is not a full reconstruction of all phonon modes but rather a histogrammed version which is an incomplete representation of the

vibrational dynamics. Fortunately, within incoherent approximation, such reduced information is sufficient for further calculation of the Debye–Waller factor using eq 2 and higher order excitations using the convolution approach already implemented in OCLIMAX following¹⁴

$$S_n(\omega) = \int_{-\infty}^{\infty} S_1(\omega - \omega') S_{n-1}(\omega') d\omega' \quad (9)$$

It should be noted that the Debye–Waller factor calculated by using eq 2 provides a larger value compared to that calculated with the use of mean-square-displacement from the MD trajectory at the same temperature. The former is a quantum treatment following the correct Bose–Einstein statistics (which includes the zero-point excitation), whereas the latter is only a classical treatment of the atoms. In fact, one can (and in some cases should) run an MD simulation at certain fictive/effective temperature that is different from the actual experimental temperature to obtain meaningful INS spectra. The fictive/effective temperature is only a means to guide the system to survey the relevant potential energy landscape, whereas the quantum phonon population, vibrational amplitude, and the Debye–Waller factor are properly accounted for through eq 2. This approach to decouple MD temperature and actual temperature can be particularly useful when studying anharmonicity or nuclear quantum effects.

RESULTS AND DISCUSSION

To demonstrate the new capability, in the following we study a few examples. The first example is the ordinary phase of water ice (ice Ih). In ice Ih, the water molecules are arranged in corrugated layers of hexagonal rings, with a stacking sequence of ABAB. The oxygen atoms occupy the vertices of the rings, and the hydrogen bonds are aligned with the edges.¹⁵ The exact positions of the hydrogen atoms are intrinsically disordered, subjected to certain rules. That is, each oxygen atom must have covalent bonds with two hydrogen atoms, forming a water molecule; it should also form two weaker hydrogen bonds with two neighboring water molecules. Similarly, each hydrogen should form one covalent bond with an oxygen atom and one hydrogen bond with another water molecule.^{16,17} A structural modeling satisfying the above rules is shown in Figure 1.

The INS spectra of ice Ih have been studied by many,^{18–27} in which the translational and librational bands exhibit rich and telltale features. Computer simulations have been performed to help interpret the features. Although the main peaks have been successfully assigned to various vibrational modes, the agreement between experiment and simulation (in terms of the peak positions and intensities) has been unsatisfactory, and it remains challenging to accurately reproduce the observed INS spectra from an atomistic model. The DFT method provides accurate calculation of interatomic forces, but it is limited to small unit cells under periodic boundary conditions, where the hydrogen disorder is inadequately represented. The conventional LD also misses the anharmonicity, which could be pronounced for the relevant vibrational modes. On the other hand, most force fields developed for water/ice are not sufficiently accurate to simulate vibrational spectra, especially in this low energy range where the details are sensitive to the hydrogen bonding and intermolecular interactions. Here we study four models of ice Ih and compare their simulated INS spectra with experiments. INS experiments were performed using the VISION²⁸ (indirect geometry) and SEQUOIA²⁹

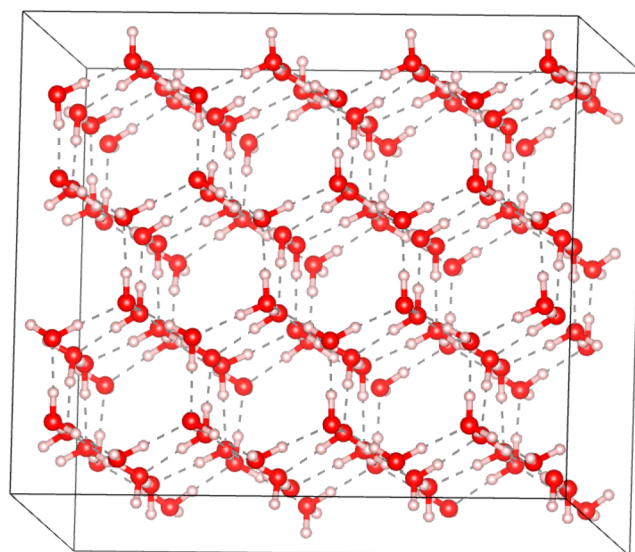


Figure 1. A structural model of ice Ih.

(direct geometry) spectrometers at the Spallation Neutron Source, Oak Ridge National Laboratory. About 0.4 g of water was loaded in an aluminum flat plate cell (about 0.1 mm thickness), and the sample was then frozen and cooled to 5 K for data collection.

We take the orthorhombic representation (unit cell $\sim 7.8 \text{ \AA} \times 7.3 \text{ \AA} \times 9.0 \text{ \AA}$) of the ice Ih crystal structure and used GenIce³⁰ to produce the initial configurations that follow the rules as previously described. The density and the a/c ratio were determined according to the experimental data at 5 K.³¹ The first model is a $2 \times 2 \times 2$ supercell containing 384 atoms (or 128 water molecules), as illustrated in Figure 1. Both LD and MD simulations were performed using the plane wave DFT code CASTEP³² (labeled DFT-222-LD and DFT-222-MD, respectively). Figure 2 shows a comparison of the

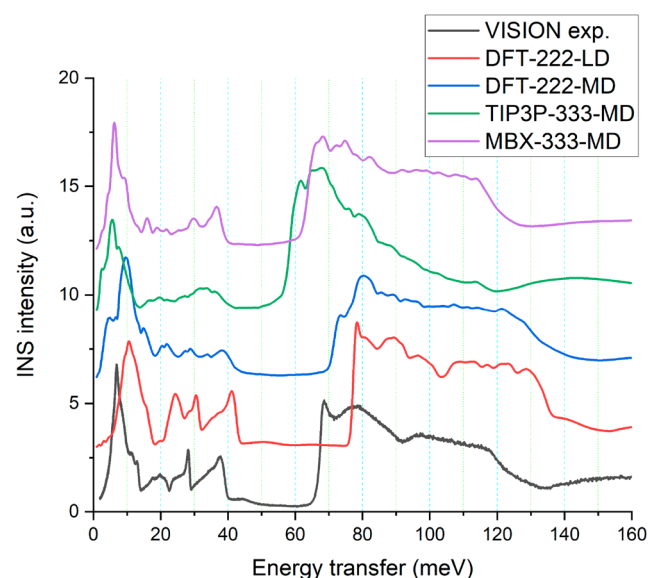


Figure 2. INS spectrum of ice Ih measured at VISION showing the translational and librational bands, in comparison with the simulated spectra from various models. The MBX-333-MD model exhibits the best agreement.

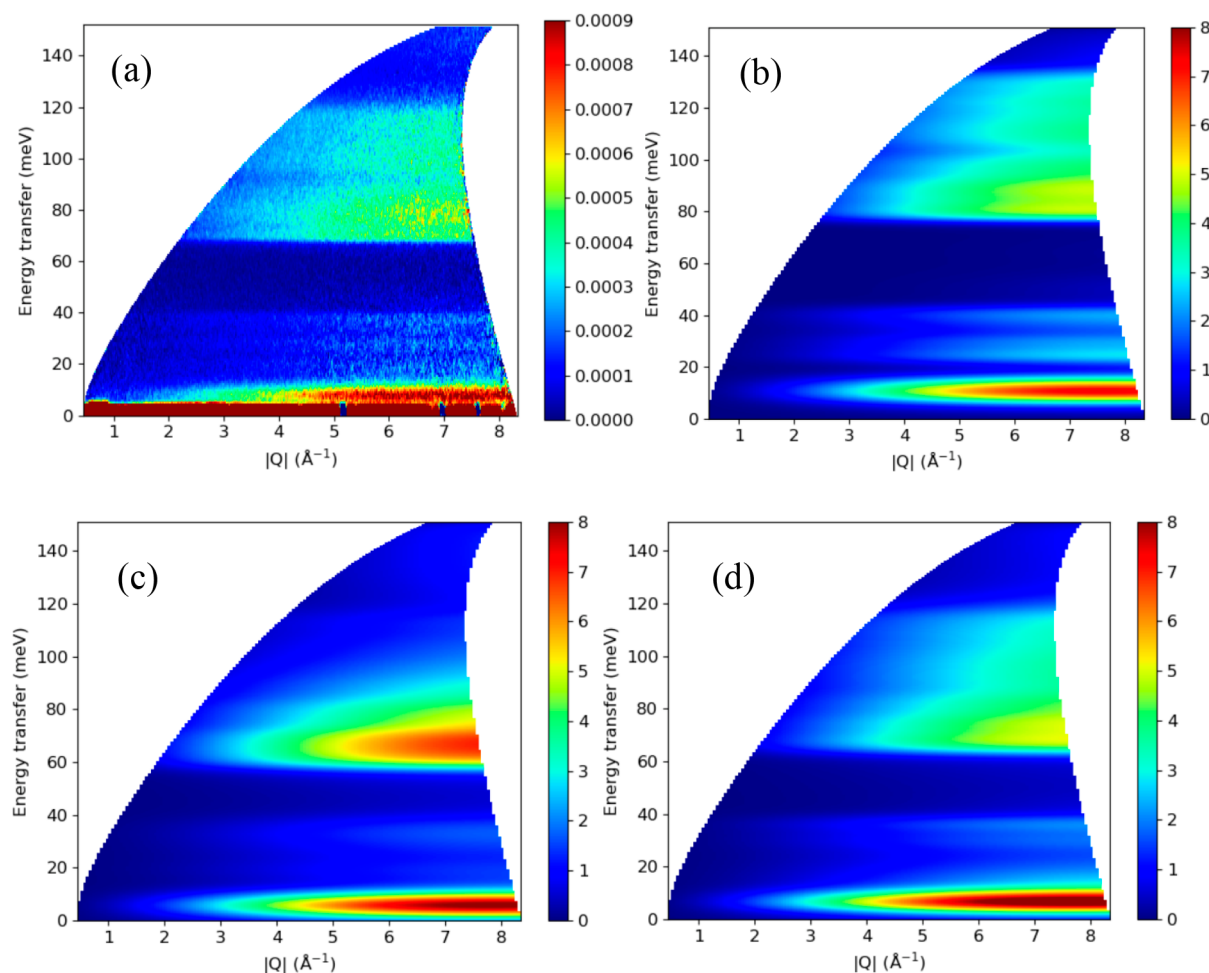


Figure 3. (a) INS spectrum of ice Ih measured at 5 K at SEQUOIA with an incident energy of 160 meV. The translational and librational bands are shown. Simulated spectra from (b) DFT-222-LD, (c) TIP3P-333-MD, and (d) MBX-333-MD models are compared with experiment.

simulations with the VISION data. It can be seen that the DFT-222-LD model reproduces the main features of the measured spectrum. However, the predicted peak positions are significantly higher, especially for the first acoustic peak and the librational band (50% and 15% discrepancy, respectively). The overestimation of the vibrational frequencies is partly due to anharmonicity of these modes in a nonparabolic local potential energy profile, with the choice of the exchange-correlation functional also playing a role.^{18,33} To directly probe the anharmonicity, an *ab initio* MD simulation of the same cell was then performed with CASTEP, and the MD trajectory was converted to INS and compared with VISION data. Indeed, all peak positions shift to lower frequencies compared to the LD simulation, confirming contribution from the anharmonicity. However, the agreement with experiment is still unsatisfactory, and then we did a classical MD simulation using the TIP3P³⁴ force field with CP2K³⁵ (labeled TIP3P-333-MD). This is a force field intended for large scale simulations of liquids and solutions, and we used a larger $3 \times 3 \times 3$ cell (1296 atoms or 432 molecules). The simulation is fast, but the accuracy is not expected to be very high. Not surprisingly, although the two main bands (translational and librational) can be recognized, the details do not agree with experiment very well. The librational modes with the TIP3P potential are also significantly softer than they should be, indicating an

insufficient description of the hydrogen bonding and intermolecular interactions. Finally, we employ the many-body water potential MB-pol^{36–39} as implemented in the MBX software⁴⁰ with i-PI⁴¹ as the MD driver to achieve the desired balance of speed and accuracy (labeled MBX-333-MD). The MB-pol has been demonstrated to be more accurate than other existing water models due to its explicit treatment of various intricate interactions and extensive comparisons with CCSD-(T) and quantum Monte Carlo reference data.⁴⁰ The simulated INS spectrum from MBX-333-MD is compared with experiment in Figure 2. The agreement is excellent in peak positions, intensities, and the overall profile of the spectra. In fact, to our knowledge, this is the best agreement that has been achieved in this low energy part of the INS spectra for ice Ih. An even larger $4 \times 4 \times 4$ cell was also studied, and no significant changes were found in the simulated spectra, meaning that convergence with respect to the cell size has been achieved.

To further verify the calculated Q dependence, we show in Figure 3 the measured and simulated spectra for SEQUOIA. In all models, the Q dependence of the intensity appears to be consistent with experiment. The DFT LD model and MBX model have a similar overall profile as the experiment, but the frequency/energy in the DFT LD model is significantly overestimated. Again, the MBX model shows the best

agreement with experiment. The favorable comparison in Figures 2 and 3 demonstrates the newly developed capability in OCLIMAX to convert MD trajectories directly to simulated INS spectra, for both direct and indirect geometry spectrometers.

Next, we would like to test the capability with a more complex molecular crystal ZIF-8 (276 atoms in a cubic unit cell of $a \sim 17$ Å).⁴² The simulated spectra from DFT LD and DFT MD are compared with experiment in Figure 4, and the

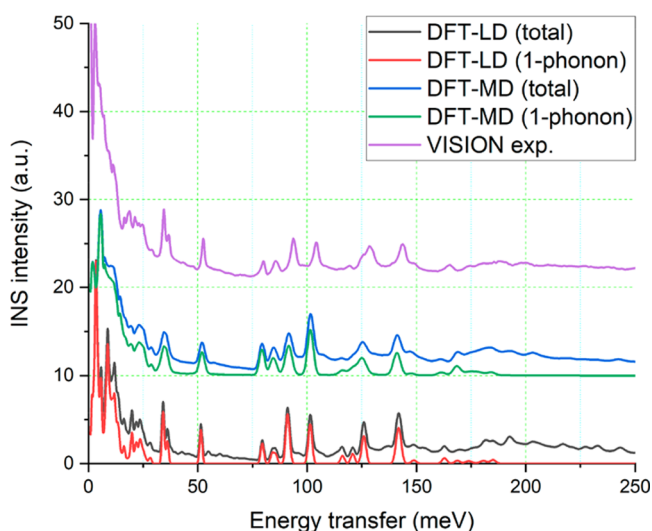


Figure 4. INS spectrum of ZIF-8 measured at VISION,⁴² in comparison to the simulated ones with DFT-LD and DFT-MD methods (offset by 10 in y-axis for clarity). Contributions from fundamental (1-phonon) excitations are shown, with the rest of the intensity coming from combinations and overtones.

agreement is excellent between the two models and between the models and the experiment. In addition, the higher order excitations (combinations and overtones) are compared, showing that the results from the MD trajectory agree with those from the LD approach, and they are responsible for the background intensities observed in experiment.

Finally, the MD approach could also be very useful to study inorganic nonhydrogenous materials, such as oxides and semiconductors. Many important inorganic solids are atomically defective, doped, disordered, or amorphous. Compared to the conventional DFT-LD approach, the MD method has obvious advantages in dealing with such complex systems. To demonstrate the capability in a representative system, Figure 5 shows an example of silica glass.⁴³ The simulation of the crystalline phase (α -quartz) with DFT-LD produces a spectrum with more and sharper peaks compared to the experiment. To make a glass with a larger cell, the CHIK potential⁴⁴ was used with the LAMMPS^{45,46} MD code. A 3000-atom cell (1000 Si and 2000 O) was melted and equilibrated at 3500 K for 10 ns. Two cooling rates were employed to make two glass samples from the melt, one with 10^9 K/s and the another with 10^8 K/s. The INS spectra for the two glass samples were calculated from MD trajectories and compared with experiment. As can be seen in Figure 5, the peaks are now much broader, consistent with the amorphous structure. The splitting peak near 140 meV (Si–O stretch) is very well reproduced. The peak at 100 meV (Si–O bending) is visible but comparably weaker. The translational peak is at 8 meV, softer than the α -quartz and in agreement with the

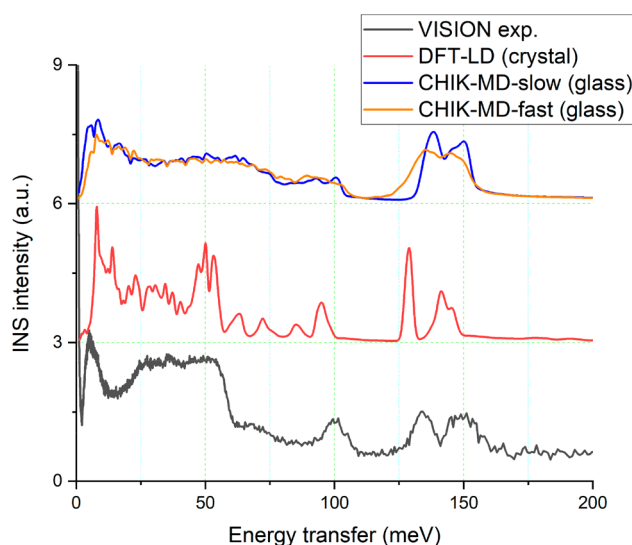


Figure 5. INS spectrum of fused silica measured at VISION,⁴³ in comparison with DFT-LD-simulated spectrum for α -quartz crystal, as well as MD-simulated spectra for silica glasses (made at two different cooling rates) using the CHIK potential.

experiment. The band between 25 and 75 meV is present but much less obvious than in experiment. We note that for all the features mentioned above, the glass sample with the slower cooling rate shows better agreement with experiment. The cooling rate used in simulation, limited by the computing power, is still at least several orders of magnitude faster than in the experiment. It is therefore expected that the agreement may be further improved if a more realistic cooling rate were used in simulation. In addition, although the CHIK potential is one of the most accurate pair potentials for silica, it still falls short in fully reproducing the phonon density of states, as already reported in the literature.^{44,47} We note that in this case, the “disagreement” is also valuable as it points us to and provides a benchmark for what we should do to improve the model. This example also further highlights the value of including INS spectra into potential validations, and the method reported in this paper will provide a key tool to enable this.

CONCLUSION

In summary, we have implemented in OCLIMAX the capability to simulate INS spectra directly from MD trajectories, complementing the existing approach based on LD. A few examples have been studied to demonstrate that the new routine can produce INS spectra reliably and accurately, for indirect and direct geometry neutron spectrometers, including not only fundamental excitations but also the combinations and overtones. The method also decouples the MD temperature and the experimental temperature so that the former is only a fictive/effective parameter to survey the potential energy landscape, while the quantum phonon population and the Debye–Waller factor are properly evaluated following Bose–Einstein statistics. The simulated INS spectra from MD trajectories are confirmed to be consistent with the traditional LD approach. This opens great opportunities for INS study of complex systems (e.g., in chemistry and biology), for which LD calculation is no longer feasible whereas MD simulations can be routinely performed with either DFT or empirical force fields. Combined with the

recently implemented capability for *ab initio* simulation of neutron scattering cross sections,⁸ the current development will also allow us to study the neutron scattering properties of complex nuclear materials using MD simulations. We believe this integration of INS and MD modeling, two widely used methods in materials research, will find important applications in many areas.

METHODS

Density Functional Theory (DFT) calculations were performed using CASTEP (version 19.11).³² The Generalized Gradient Approximation (GGA), as implemented by Perdew–Burke–Ernzerhof (PBE),⁴⁸ was used to describe the exchange–correlation interactions. On-the-fly ultrasoft pseudopotentials were employed to account for the effects of core electrons, with the default energy cutoff for plane-wave basis. The atomic coordinates were relaxed to allow minimization of the potential energy and the interatomic forces. The energy tolerance for the electronic structure calculations was 5×10^{-10} eV, and the energy tolerance for ionic relaxation was 5×10^{-9} eV. The tolerance for the interatomic forces was 1 meV/Å. After convergence was reached, the dynamical matrix was obtained using the finite displacement method, from which the phonon frequencies and vibrational modes were calculated.

For MD simulations, CASTEP was used to run all DFT MD, CP2K³⁵ was used with the TIP3P³⁴ potential for ice, MBX⁴⁰ and i-PI⁴¹ were used for the MB-pol^{36–38} ice model, and LAMMPS^{45,46} was used with the CHIK⁴⁴ potential for silica. The MD simulation time-step was 0.2 fs for ice, 0.5 fs for ZIF-8, and 1.0 fs for silica. The MD temperature for ice and ZIF-8 was 180 K. This intermediate temperature was chosen so that the quantum zero-point fluctuation of protons is partially and approximately accounted for by a classical thermal fluctuation. As stated in the paper, this temperature was only used to survey the potential energy landscape, and the quantum Bose–Einstein statistics was used to calculate the actual phonon population and the Debye–Waller factor. For silica, the system was first melted at 3500 K for 10^7 time-steps. It was then quenched to 300 K at two different rates, 10^9 K/s and 10^8 K/s. For all other MD simulations, the potential energy of the initial structure was first minimized, and then the system was equilibrated under the NVT ensemble at the designated temperature for about 5000 time-steps. Finally, an NVE trajectory of 50000 time-steps for ice (20000 for ZIF-8 and 10000 for silica) was generated for INS simulation. To convert MD trajectories to INS spectra, OCLIMAX can be used in essentially the same way as reported previously, with the same format for the parameter file.⁷ Here instead of having the aclimax or oclimax file for phonon information, we need a tclimax file for the trajectory. The tclimax file can be obtained from the MD output of various programs such as CASTEP, CP2K, LAMMPS, etc. using an included format-converting script.

AUTHOR INFORMATION

Corresponding Author

Y. Q. Cheng – Neutron Scattering Division, Oak Ridge National Laboratory, Oak Ridge, Tennessee 37831, United States; orcid.org/0000-0002-3263-4812; Email: chengy@ornl.gov

Authors

A. I. Kolesnikov – Neutron Scattering Division, Oak Ridge National Laboratory, Oak Ridge, Tennessee 37831, United States; orcid.org/0000-0003-1940-4649

A. J. Ramirez-Cuesta – Neutron Scattering Division, Oak Ridge National Laboratory, Oak Ridge, Tennessee 37831, United States; orcid.org/0000-0003-1231-0068

Complete contact information is available at:

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Notes

The authors declare no competing financial interest.

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